



Institut Universitaire des sciences

- Faculté des sciences et de technologie
- TD N°3 Réseau II

Nom & Prénom : COFFY Cliford

Niveau : L3

Prof. : Ismael SAINT AMOUR

Date : 02/04/2026

Sécurité réseau

L'objectif de ce TD est de :

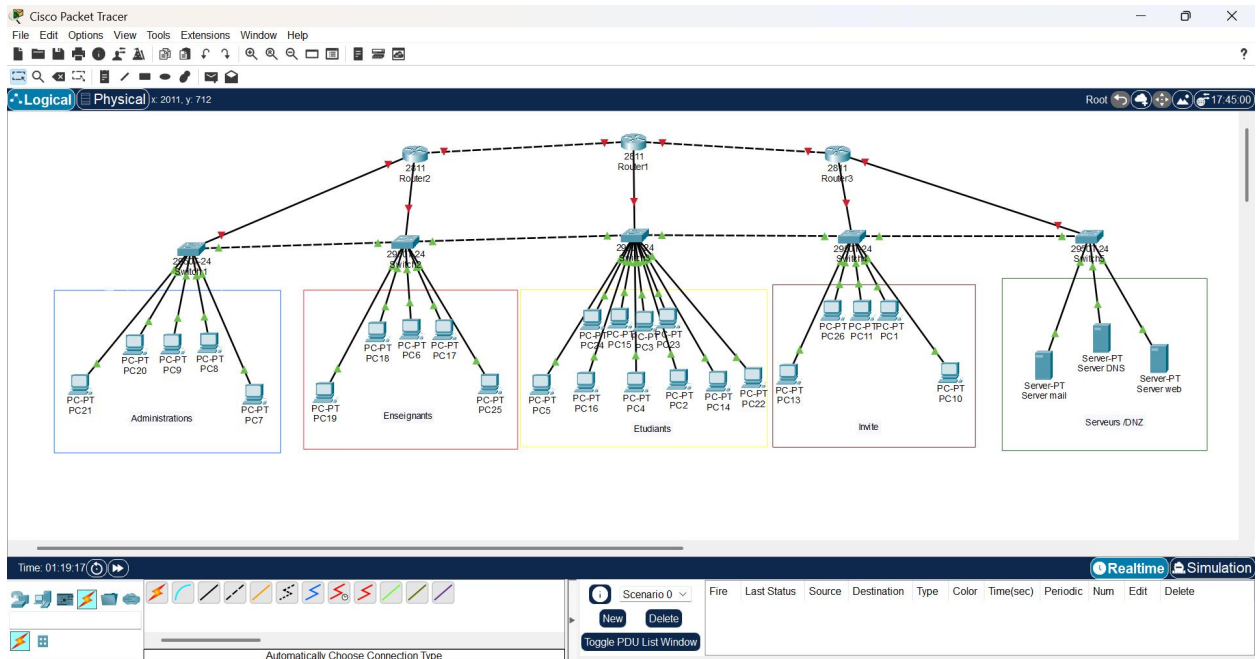
1. Apprendre à créer et gérer plusieurs VLANs pour isoler les différents départements (Administration, Commercial, Technique, Invités, DMZ).
2. Comprendre le routage inter-VLAN à l'aide de routeurs et de sous-interfaces.
3. Identifier les avantages de la segmentation pour la sécurité.
4. Configurer des mots de passe forts pour l'accès console et VTY.
5. Activer SSH sur les routeurs pour un accès sécurisé à distance.
6. Appliquer la port-security sur les switches pour limiter les adresses MAC autorisées sur chaque port.
7. Créer des ACL étendues pour contrôler l'accès entre VLANs (ex. : VLAN invités ne peuvent pas accéder à la DMZ ou aux ressources sensibles).
8. Filtrer le trafic réseau de façon sélective (protocole, IP source/destination).
9. Simulation de scénarios d'attaque et test

Travaux Dirigés

Sécurisé le réseau sur un campus universitaire avec Cisco Packet Tracer.

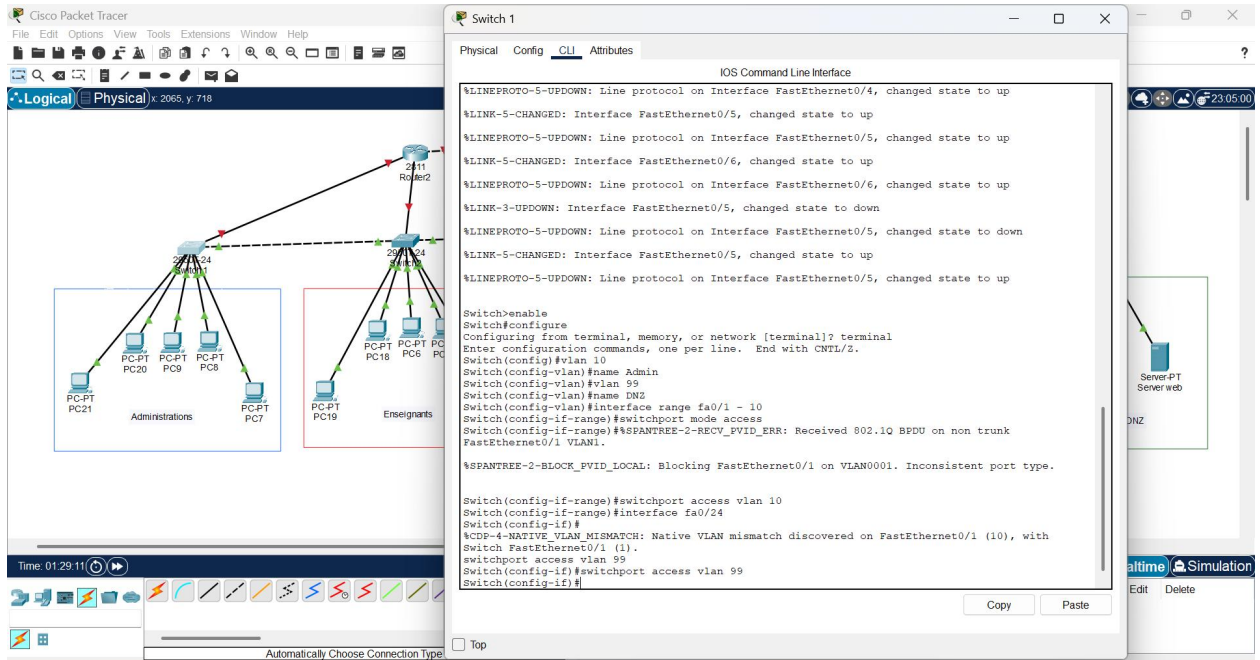
1. Création de la topologie réseau

Ici vous pouvez donc voir la création du topologie avec 3 routeurs, 5 switches et 25 Pcs et les serveurs en connection



2. Configurer les VLANs

➤ Configuration du VLANs du Switch de L'administration :



The image shows a Cisco Packet Tracer network diagram and the CLI configuration for Switch 1. The network diagram on the left shows a central switch (Switch 1) connected to a router (Router 2) and two groups of PCs: Administrations (PC20, PC21, PC22, PC23, PC24) and Enseignants (PC18, PC19, PC20, PC21, PC22, PC23, PC24). The CLI window on the right shows the configuration for Switch 1, including enabling the switch, configuring the terminal, and setting up VLANs 10 and 99. The configuration also includes interface settings for FastEthernet0/1, FastEthernet0/24, and FastEthernet0/25.

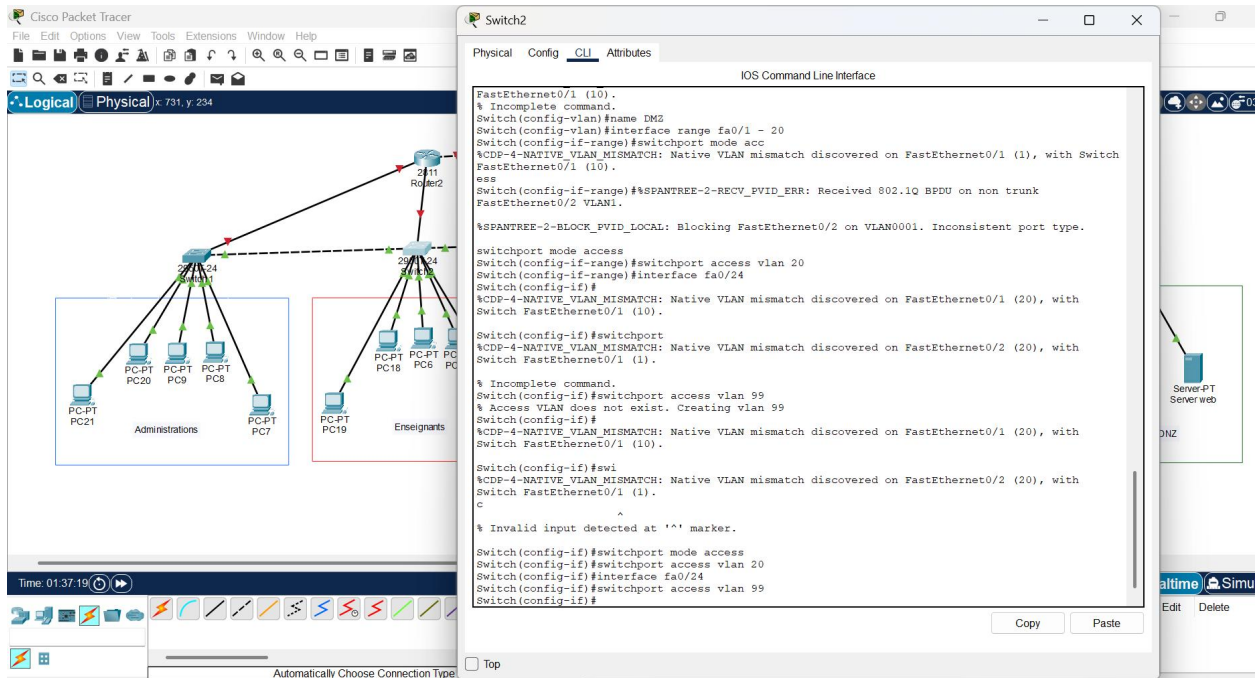
```
Switch 1
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch>enable
Switch#configure
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Admin
Switch(config-vlan)#vlan 99
Switch(config-vlan)#interface range fa0/1 - 10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk
FastEthernet0/1 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN0001. Inconsistent port type.

Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#interface fa0/24
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with
Switch FastEthernet0/1 (1).
switchport access vlan 99
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
```

➤ Configuration du VLANs du Switch des Enseignants :



The image shows a Cisco Packet Tracer network diagram and the CLI configuration for Switch 2. The network diagram on the left shows a central switch (Switch 2) connected to a router (Router 2) and two groups of PCs: Administrations (PC20, PC21, PC22, PC23, PC24) and Enseignants (PC18, PC19, PC20, PC21, PC22, PC23, PC24). The CLI window on the right shows the configuration for Switch 2, including enabling the switch, configuring the terminal, and setting up VLANs 10 and 99. The configuration also includes interface settings for FastEthernet0/1, FastEthernet0/24, and FastEthernet0/25.

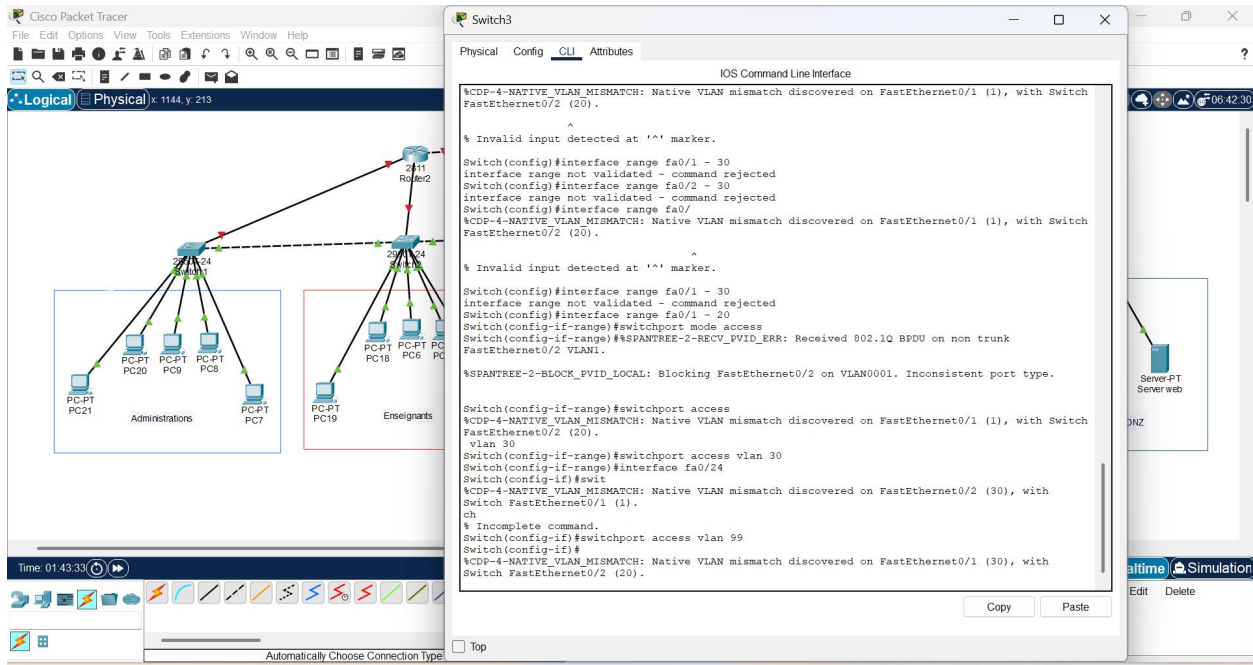
```
Switch 2
IOS Command Line Interface

FastEthernet0/1 (10).
% Incomplete command.
Switch(config-vlan)#name DM2
Switch(config-vlan)#interface range fa0/1 - 20
Switch(config-if-range)#switchport mode acc
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/1 (10).
ess
Switch(config-if-range)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk
FastEthernet0/2 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent port type.

switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#interface fa0/24
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (20), with
Switch FastEthernet0/1 (10).
Switch(config-if)#switchport
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (20), with
Switch FastEthernet0/1 (1).
% Incomplete command.
Switch(config-if)#switchport access vlan 99
% Access VLAN does not exist. Creating vlan 99
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (20), with
Switch FastEthernet0/1 (10).
Switch(config-if)#swi
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (20), with
Switch FastEthernet0/1 (1).
c
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface fa0/24
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
```

➤ Configuration du VLANs du Switch des Etudiants :



Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/2 (20).
```

```
% Invalid input detected at '^' marker.
```

```
Switch(config)#interface range fa0/1 - 30
interface range not validated - command rejected
Switch(config)#interface range fa0/2 - 30
interface range not validated - command rejected
Switch(config)#interface range fa0/
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/2 (20).
```

```
% Invalid input detected at '^' marker.
```

```
Switch(config)#interface range fa0/1 - 30
interface range not validated - command rejected
Switch(config)#interface range fa0/1 - 20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/2 VLAN1.
```

```
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent port type.
```

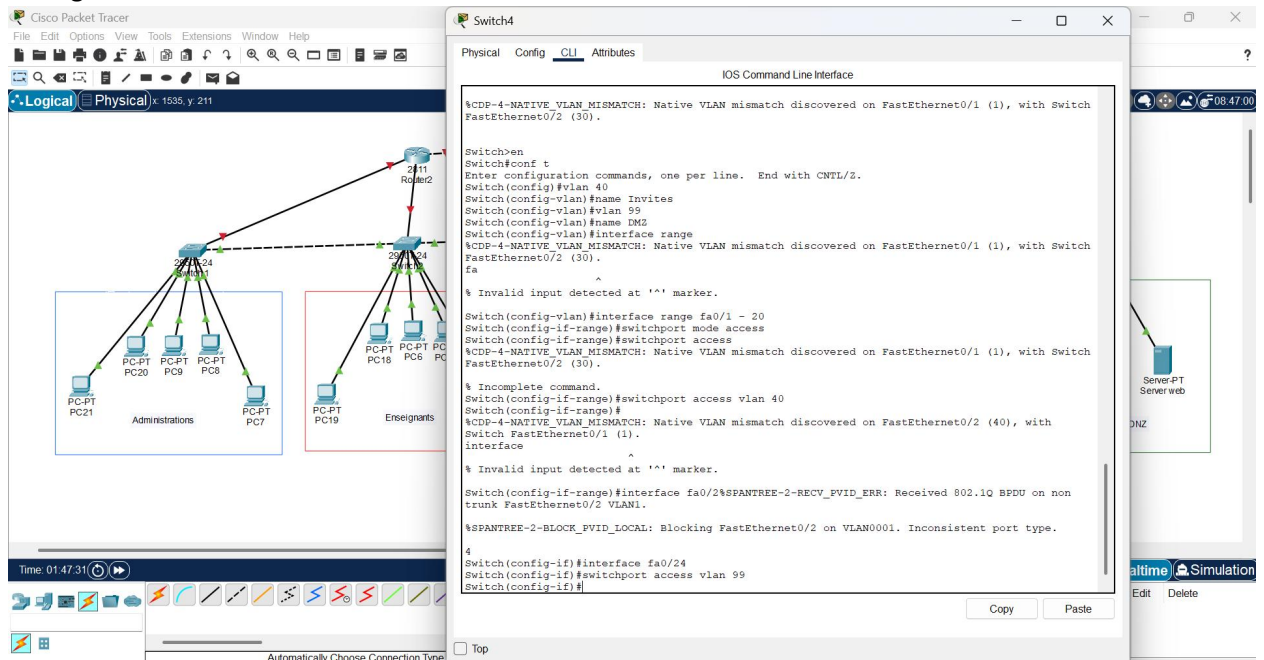
```
Switch(config-if-range)#switchport access
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/2 (20).
```

```
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#interface fa0/24
Switch(config-if)#swit
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (30), with Switch FastEthernet0/1 (1).
```

```
ch
% Incomplete command.
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (30), with Switch FastEthernet0/2 (20).
```

Copy Paste

➤ Configuration du VLANs du Switch des Invités :



Switch4

Physical Config CLI Attributes

IOS Command Line Interface

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/2 (30).
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 40
Switch(config-vlan)#name Invites
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name DMZ
Switch(config-vlan)#interface range
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/2 (30).
```

```
% Invalid input detected at '^' marker.
```

```
Switch(config-vlan)#interface range fa0/1 - 20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/2 (30).
```

```
% Incomplete command.
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (40), with Switch FastEthernet0/1 (1).
```

```
% Invalid input detected at '^' marker.
```

```
Switch(config-if-range)#interface fa0/2%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/2 VLAN1.
```

```
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent port type.
```

```
4
Switch(config-if)#interface fa0/24
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
```

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➤ Configuration du VLANs du Switch des Serveurs /DMZ:

The image displays a Cisco Packet Tracer simulation environment. On the left, a network diagram shows a central switch (Switch5) connected to a router (Router2). The switch is configured with two VLANs: 'Administrations' (VLAN 99) and 'Enseignants' (VLAN 40). The 'Administrations' VLAN includes PCs PC21, PC20, PC9, PC8, and PC7. The 'Enseignants' VLAN includes PCs PC19, PC18, PC6, and PC5. The router is connected to the switch via its Fa0/24 interface. On the right, the CLI window for Switch5 shows the configuration process. The configuration includes setting the switchport mode to access, creating VLAN 99, and assigning the Fa0/24 interface to VLAN 99. The CLI also displays several status messages, including 'LINEPROTO-5-UPDOWN' and 'CDP-4-NATIVE_VLAN_MISMATCH'.

Switch5

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (40).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (40).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (40).

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 99
Switch(config-vlan)#name DMZ
Switch(config-vlan)#interface range fa0/1 - 2
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (40).
0
Switch(config-if-range)#interface range fa0/1 - 20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#interface fa0/24
Switch(config-if)#interface fa0/24
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch
FastEthernet0/2 (40).

Switch(config-if)#switchport access vlan 99
Switch(config-if)#
```

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Time: 01:51:18

Automatically Choose Connection Type

3. Configurer le routage inter-VLAN

- Sur les routeurs, créer des sous-interfaces VLAN avec des adresses IP pour permettre la communication entre VLANs autorisés.
- Vérifier la connectivité avec ping

Premier router pour vlan 10

Router4

IOS Command Line Interface

to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at: <http://www.cisco.com/wml/export/crypto/tool/stqrg.html>

If you require further assistance please contact us by sending email to export@cisco.com.

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0.10
% Invalid input detected at '^' marker.
Router(config)#interface fa0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#interface fa0/0.99
Router(config-subif)#encapsulation dot1Q 99
% Invalid input detected at '^' marker.
Router(config-subif)#encapsulation dot1Q 99
Router(config-subif)#ip address 192.168.99.1 255.255.255.0
Router(config-subif)#
```

Router4

Device Name: Router4
Custom Device Model: 2811 IOS15
Hostname: Router

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/0	Down	--	<not set>	<not set>	000C.853C.2301
FastEthernet0/0.10	Down	--	192.168.10.1/24	<not set>	000C.853C.2301
FastEthernet0/0.99	Down	--	192.168.99.1/24	<not set>	000C.853C.2301
FastEthernet0/1	Down	--	<not set>	<not set>	000C.853C.2302
Vlan1	Down	1	<not set>	<not set>	0001.6309.4E28

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router4

Administrations: PC-PT PC21, PC-PT PC20, PC-PT PC19, PC-PT PC18, PC-PT PC17, PC-PT PC16, PC-PT PC15, PC-PT PC14, PC-PT PC13, PC-PT PC12, PC-PT PC11, PC-PT PC10

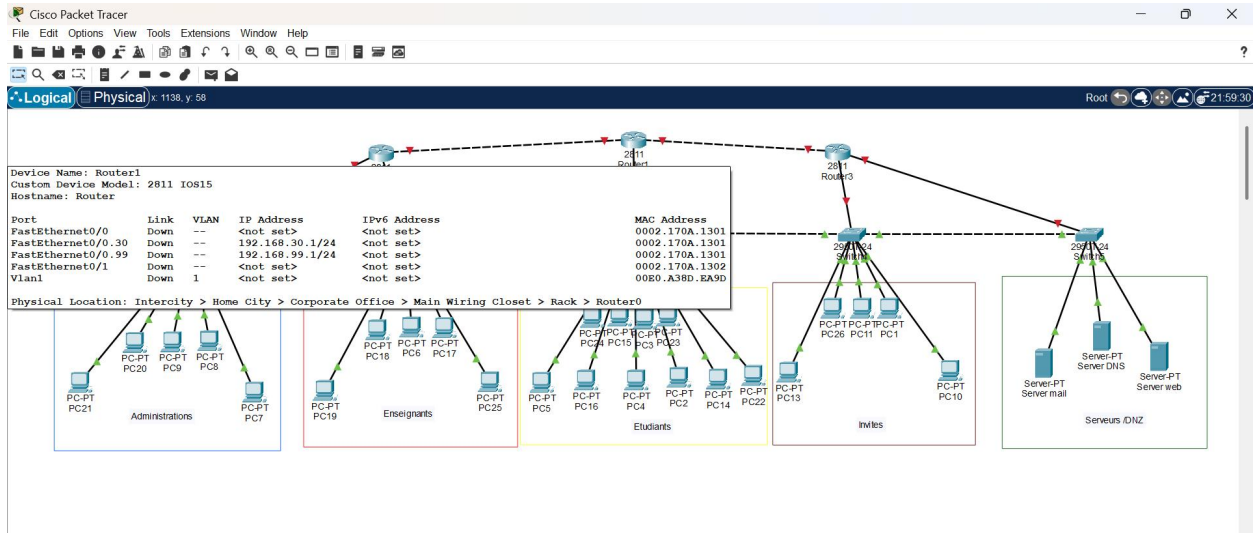
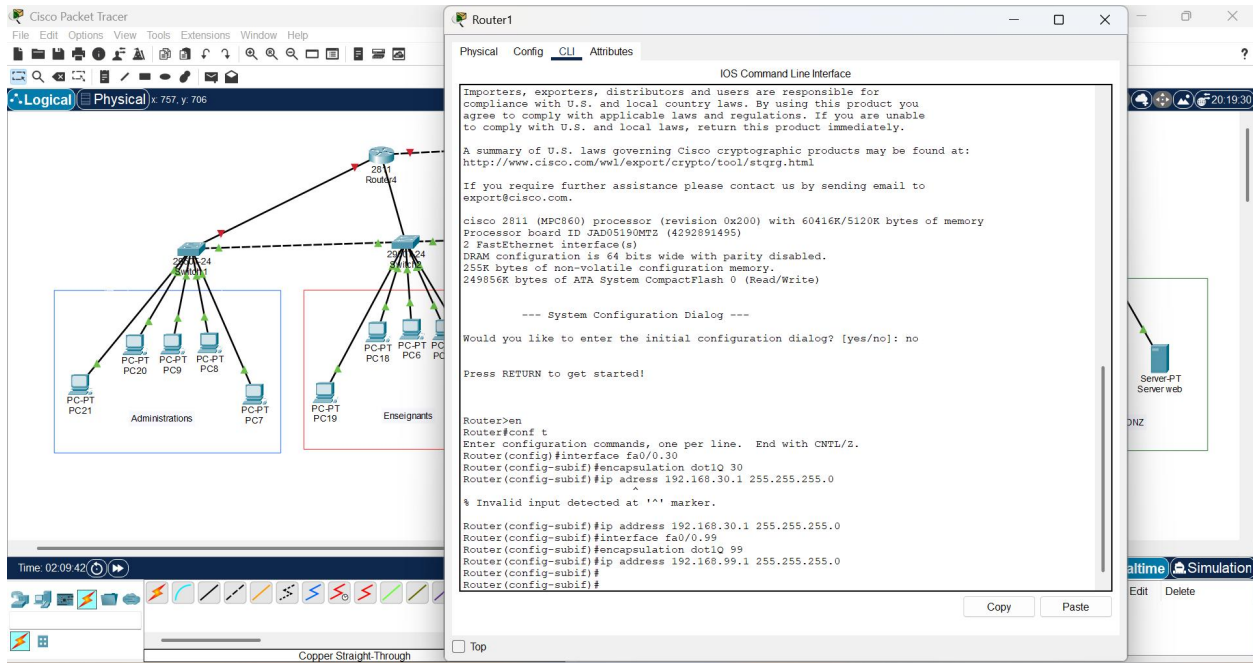
Enseignants: PC-PT PC19, PC-PT PC18, PC-PT PC17, PC-PT PC16, PC-PT PC15, PC-PT PC14, PC-PT PC13, PC-PT PC12, PC-PT PC11, PC-PT PC10

Etudiants: PC-PT PC19, PC-PT PC18, PC-PT PC17, PC-PT PC16, PC-PT PC15, PC-PT PC14, PC-PT PC13, PC-PT PC12, PC-PT PC11, PC-PT PC10

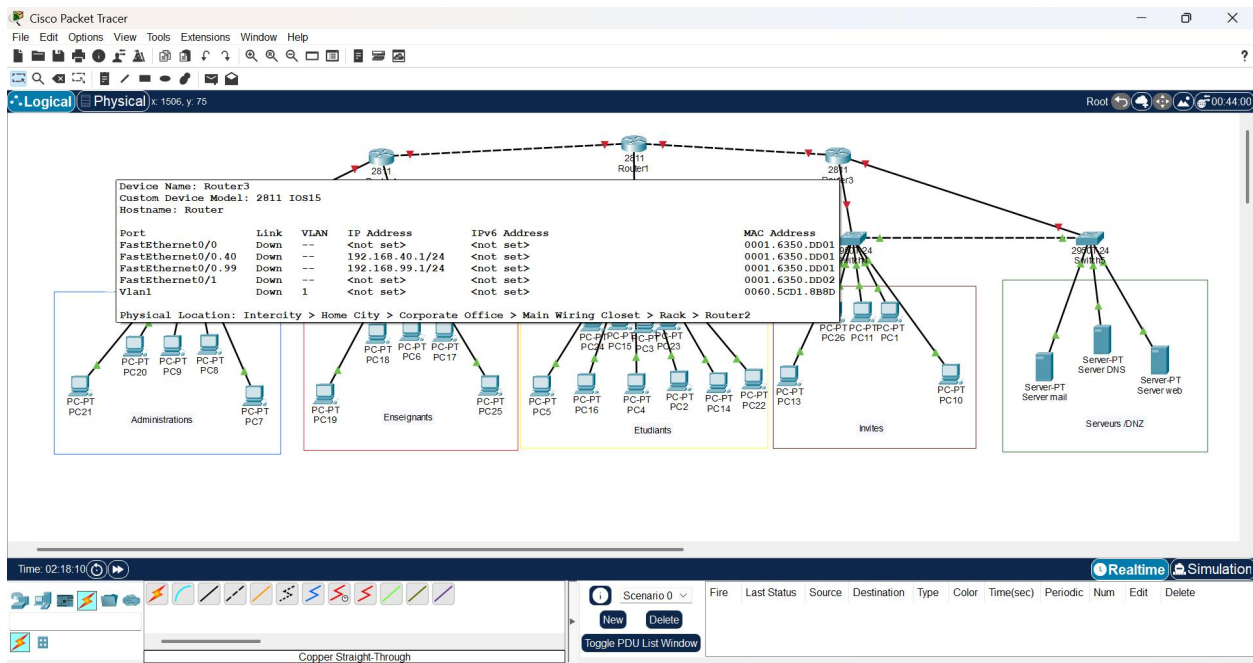
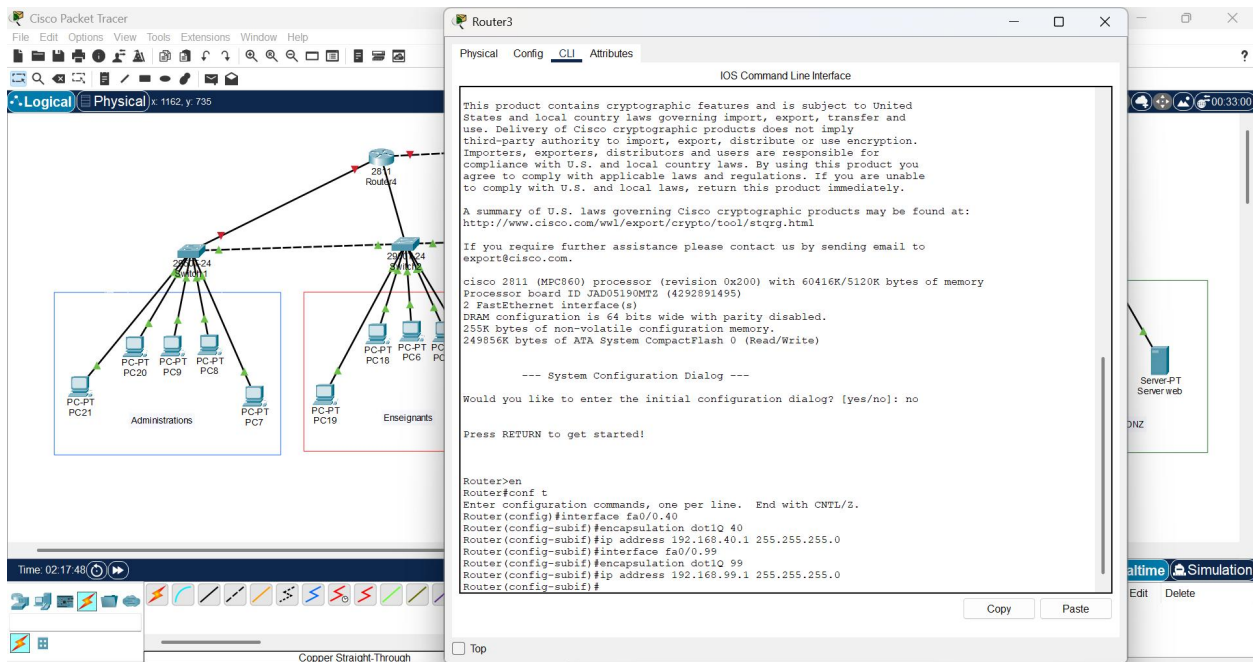
Invites: PC-PT PC19, PC-PT PC18, PC-PT PC17, PC-PT PC16, PC-PT PC15, PC-PT PC14, PC-PT PC13, PC-PT PC12, PC-PT PC11, PC-PT PC10

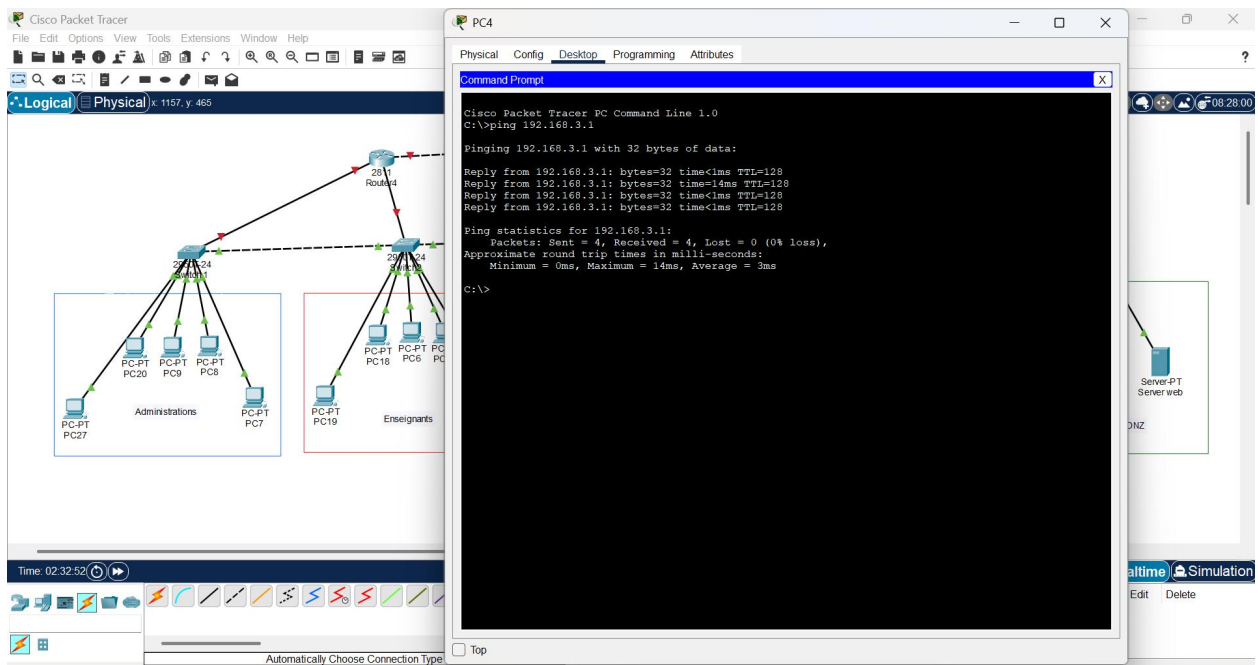
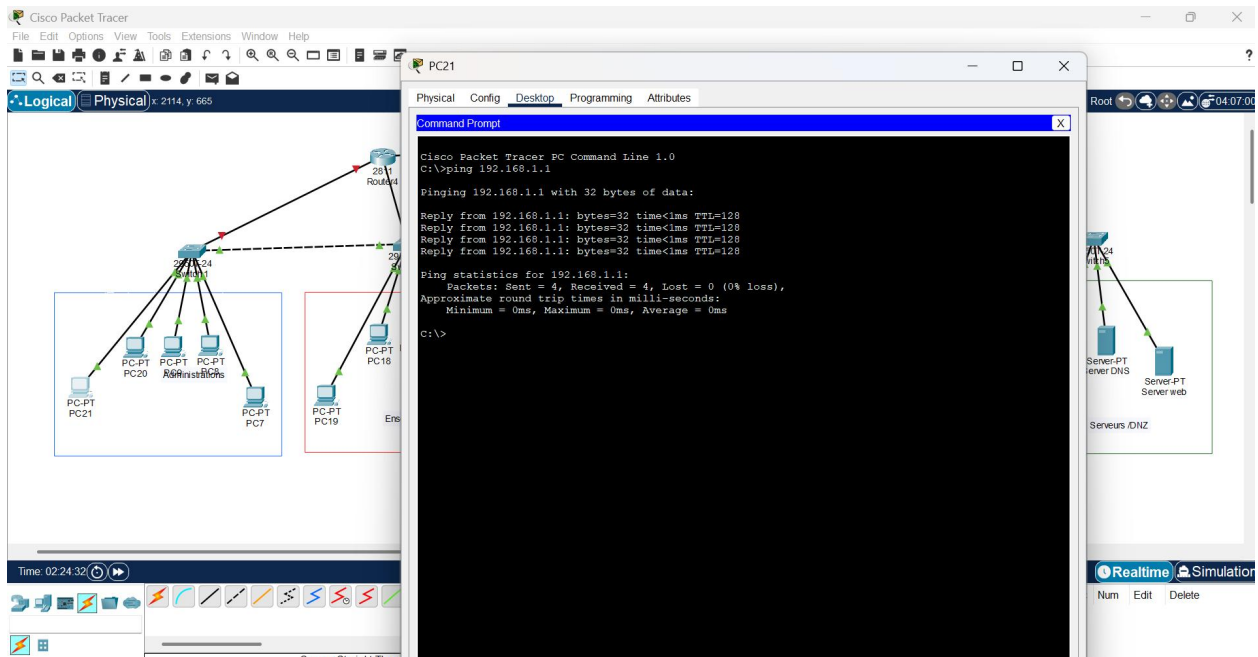
Serveurs: Server-PT Server mail, Server-PT Server DNS, Server-PT Server web

Deuxieme router associe à leurs vlans :



Troisième router associe à leurs VLANs :





Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x 1917, y 670

PC10

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.1

Pinging 192.168.4.1 with 32 bytes of data:

Reply from 192.168.4.1: bytes=32 time<1ms TTL=128
Reply from 192.168.4.1: bytes=32 time=5ms TTL=128
Reply from 192.168.4.1: bytes=32 time=1ms TTL=128
Reply from 192.168.4.1: bytes=32 time=13ms TTL=128

Ping statistics for 192.168.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 5ms
C:\>
```

Time: 02:35:32

Automaticall...

Root

Server-PT Server mail

Server-PT Server DNS

Server-PT Server web

Serveurs DNS

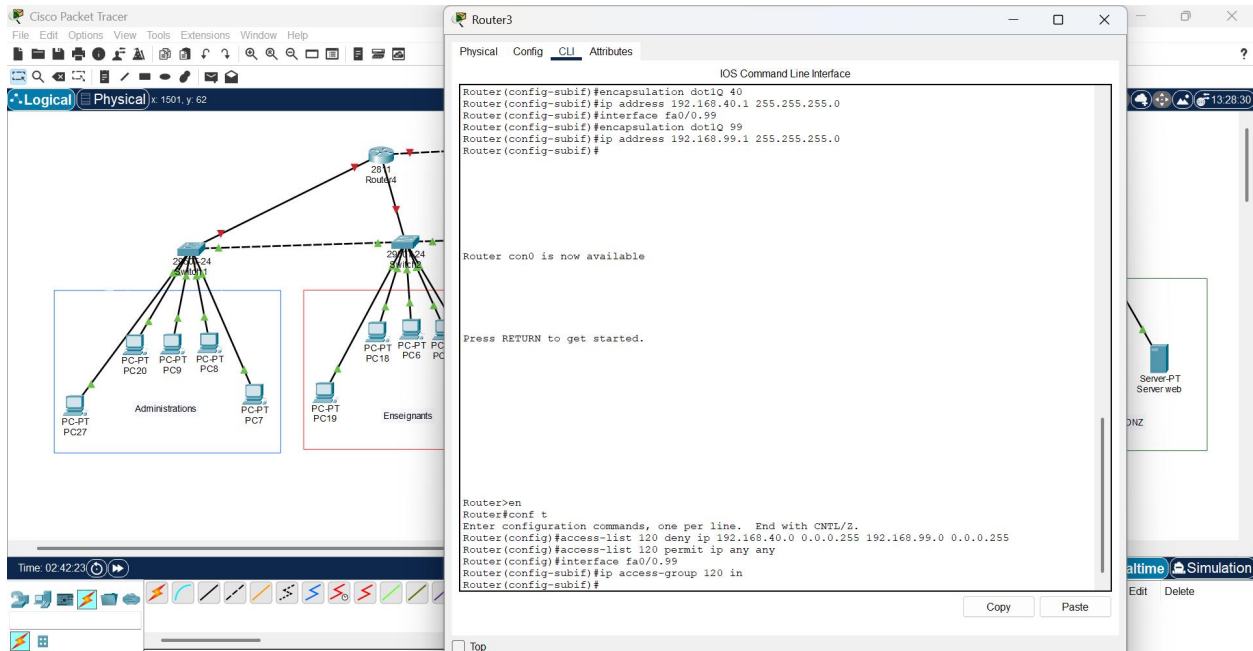
Realtime Simulation

pc) Periodic Num Edit Delete

4. Mettre en place la sécurité réseau*

1. ACL (Access Control List)

- ✓ Ici vous pouvez voir toutes les configurations pour bloquer certains VLAN (ex. invités) d'accéder à la DMZ ou à l'administration et de Permettre aux VLAN autorisés de communiquer avec les serveurs.



2. Port Security

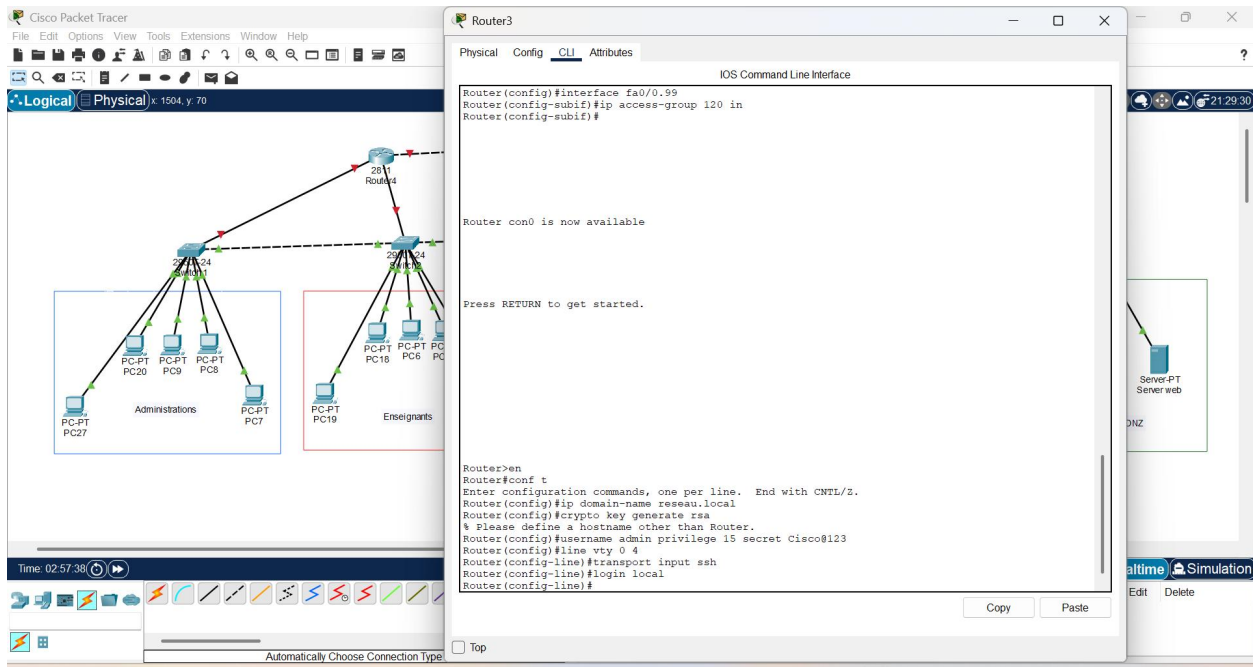
- Limiter le nombre d'appareils par port sur les switches.
- Configurer la violation pour restreindre l'accès si un PC non autorisé se connecte.

The image shows a Cisco Packet Tracer network diagram and the CLI configuration for Switch4. The network diagram on the left shows a central switch (Switch4) connected to a router (R1) and two groups of PCs: Administrations (PC20, PC9, PC8, PC7) and Enseignants (PC18, PC6, PC5, PC19). The router is connected to the switch via its 2611 interface. The switch is connected to the PCs via its 24 interfaces. The CLI configuration on the right shows the configuration for Switch4, including the configuration of port security on interfaces FastEthernet0/21, 22, 23, 24, and 30. The configuration includes commands for setting the maximum number of MAC addresses, the violation mode (restrict), and the sticky MAC address learning. The configuration also includes a command to set the native VLAN mismatch discovery on FastEthernet0/1 (40).

```
IOS Command Line Interface
Command rejected: FastEthernet0/21 is a dynamic port.
Command rejected: FastEthernet0/22 is a dynamic port.
Command rejected: FastEthernet0/23 is a dynamic port.
Command rejected: FastEthernet0/24 is a dynamic port.
Switch(config-if-range)#switchport port-security
Command rejected: FastEthernet0/21 is a dynamic port.
Command rejected: FastEthernet0/22 is a dynamic port.
Command rejected: FastEthernet0/23 is a dynamic port.
Command rejected: FastEthernet0/24 is a dynamic port.
Switch(config-if-range)#switchport port-security maximum 2
Switch(config-if-range)#switchport port-security violation restr
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (40), with
Switch FastEthernet0/2 (30).
^
% Invalid input detected at '^' marker.
Switch(config-if-range)#switchport port-security violation restrict
Switch(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (40), with
Switch FastEthernet0/1 (1).
switchport port-security violation mac-address sticky
^
% Invalid input detected at '^' marker.
Switch(config-if-range)#switchport port-security violation mac-address sticky
^
% Invalid input detected at '^' marker.
Switch(config-if-range)#switchport port-security violation mac-address sticky
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (40), with
Switch FastEthernet0/2 (30).
^
% Invalid input detected at '^' marker.
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (40), with
Switch FastEthernet0/1 (1).
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if-range)#
Switch(config-if-range)#
```

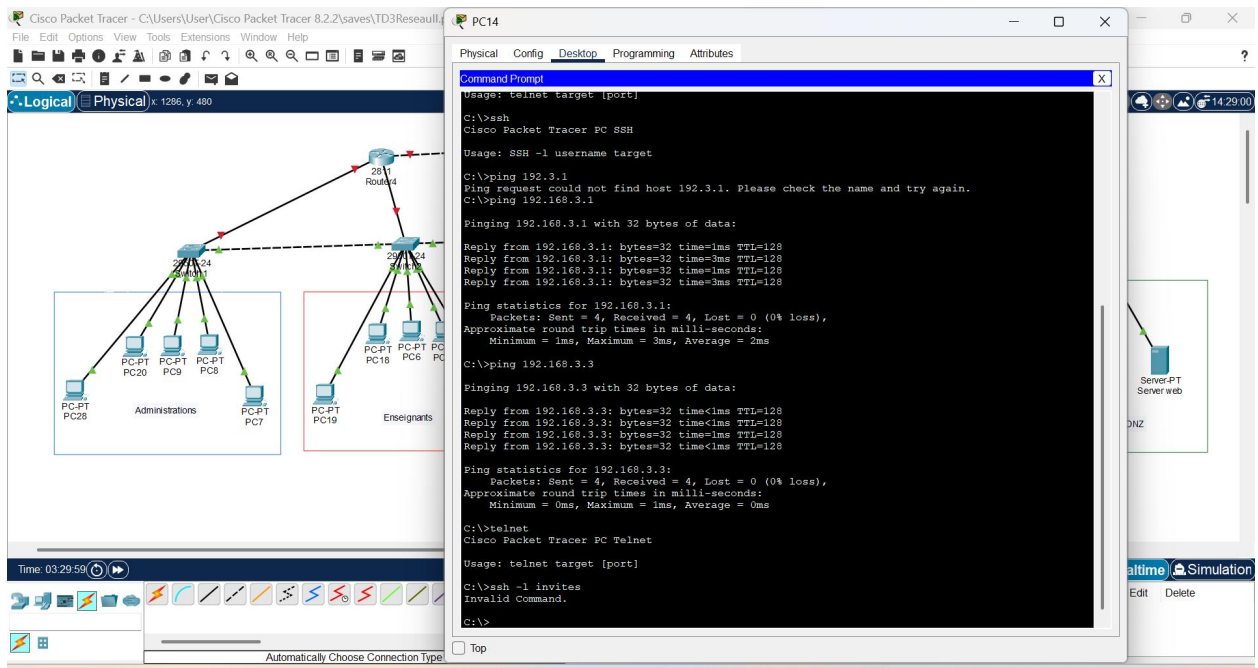
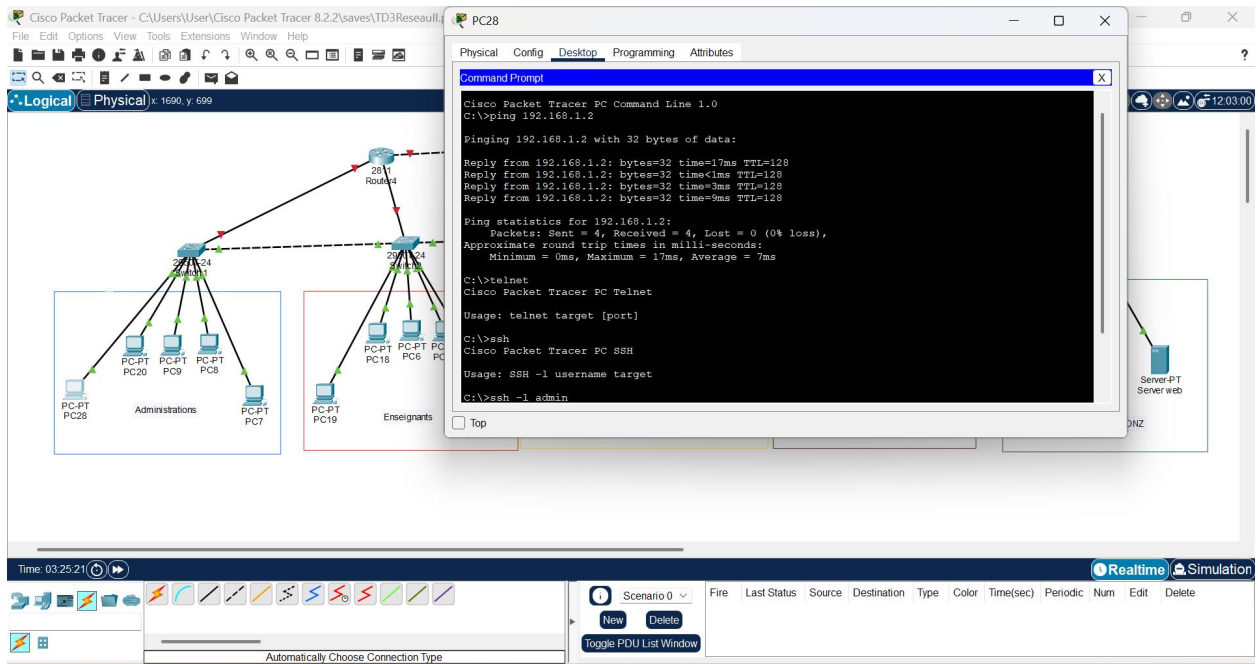
3. Sécurisation des routeurs

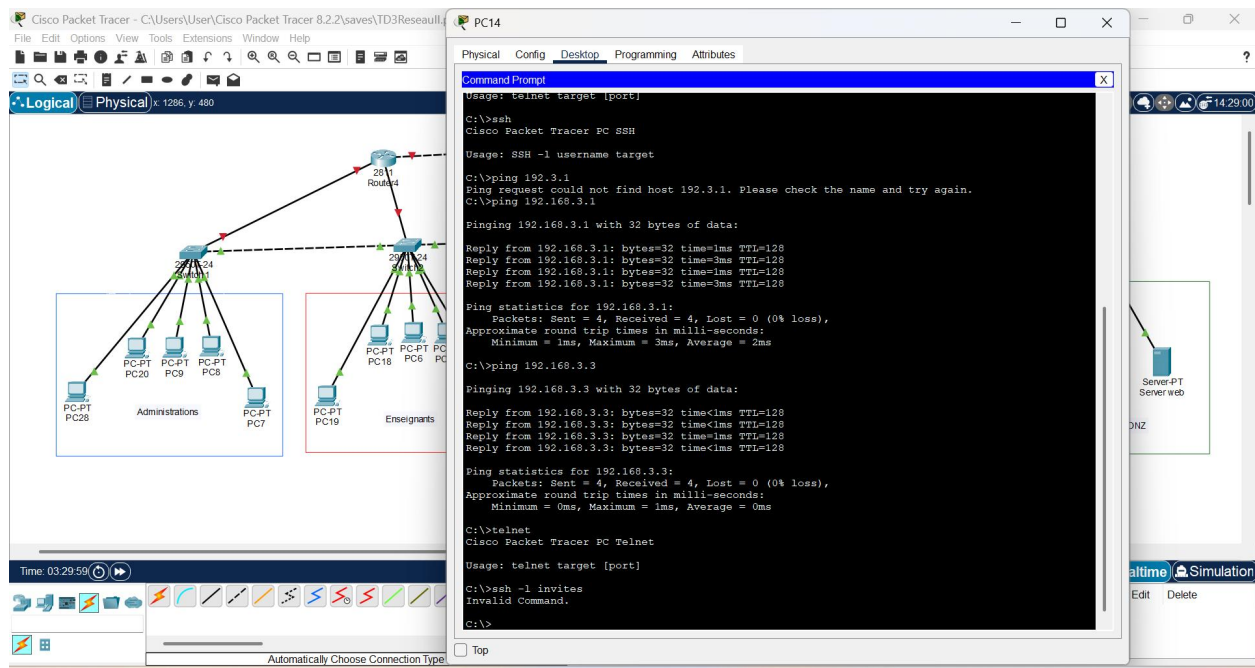
- Désactiver Telnet, activer SSH pour l'administration.
- Configurer des mots de passe forts.



Scénarios de test pour le TD

1. Ping entre VLAN autorisés : admin → enseignants → doit fonctionner.
2. Ping VLAN invités → serveur DMZ : doit échouer.
3. SSH routeurs : tester que Telnet est désactivé et SSH fonctionne.
4. Port Security : connecter un 3e PC sur un port limité → doit être restreint.
5. Accès aux serveurs : seuls admin et enseignants autorisés.





Objectifs du TD

1. **Conception de la topologie réseau**
 - Apprendre à organiser les routeurs, switches et postes selon les rôles (administration, enseignants, étudiants, invités, serveurs).
 - Comprendre la segmentation logique du réseau.
2. **Configuration des VLANs**
 - Mettre en œuvre la séparation des domaines de diffusion.
 - Appliquer une logique de sécurité et de performance.
3. **Routing inter-VLAN**
 - Permettre la communication sélective entre VLANs via des sous-interfaces.
 - Maîtriser le routage sur un routeur Cisco.
4. **Mise en place de la sécurité**
 - Utiliser les **ACL** pour filtrer le trafic entre VLANs.
 - Appliquer la **Port Security** pour limiter les connexions physiques.
 - Sécuriser les routeurs avec **SSH** et des mots de passe forts.
5. **Validation et tests**
 - Vérifier la connectivité autorisée et bloquée (ping, SSH, accès serveur).
 - Simuler des scénarios de violation pour observer les réactions du réseau.